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- Lesson Plan - Group
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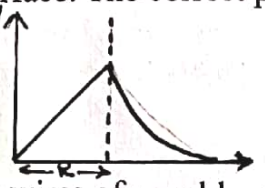
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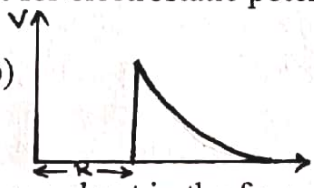
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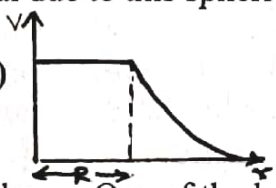
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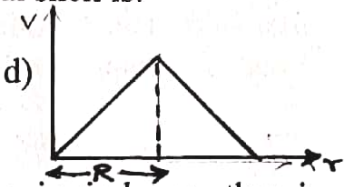
Questions and Answers - PTA - MODEL QUESTION PAPER -1

1. A thin conducting spherical shell of radius R has a charge Q which is uniformly distributed on its surface. The correct plot for electrostatic potential due to this spherical shell is:

a) 

b) 

c) 

d) 

2. Three wires of equal lengths are bent in the form of loops. One of the loop is circle , another is a semicircle and the third one is a square. They are placed in a uniform magnetic field and the same current passed through them. Which of the following loop configuration will experience greater torque?

a) circle b) semi – circle c) square d) all of them

Hints: $\tau = BA\sin\theta$ $\tau \propto A$; $A_{circle} > A_{semi\ circle} > A_{square}$ hence $\tau_{circle} > \tau_{semi\ circle} > \tau_{square}$

3. The particle which gives mass to protons and neutrons are

a) Bulk particle **b) Higgs particle** c) Nano particle d) Einstein particle

4. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom is a) 1 : -2 **b) 1 : -1** c) 1 : 1 d) 2 : -1

5. A device that converts variations in a physical quantity (pressure, temperature, sound) into an equivalent electrical signal is called:

a) modulator b) oscillator **c) transducer** d) demodulator

6. A 800 – turn coil of effective area 0.05 m^2 is kept perpendicular to a magnetic field 5×10^{-5} T. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s, the emf induced in the coil will be

a) 2 V b) 0.2 V c) 0.002 V **d) 0.02**

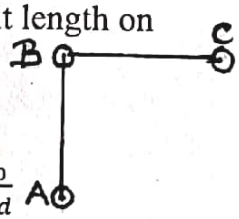
Hints: $\varepsilon = \frac{N(\phi_2-\phi_1)}{t} = \frac{NBA\cos\phi_2 - NBA\cos\phi_1}{t} = \frac{NBA\cos 0 - NBA\cos 90}{t}$

$= \frac{NBA}{t} = \frac{800\times5\times10^{-5}\times5\times10^{-2}}{0.1} = 0.02$ V

7. Name of the EM waves used for studying molecular structure and also in forensic investigations

a) UV rays b) Gamma rays c) X – rays d) IR rays

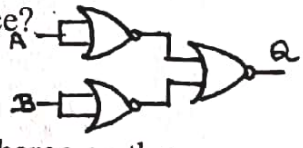
8. An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current I along the same direction is shown in fig. Magnitude of force per unit length on the middle wire B is given by



a) $\frac{2I^2\mu_0}{\pi d}$ **b)** $\frac{I^2\mu_0}{\sqrt{2}\pi d}$ c) $\frac{\sqrt{2}I^2\mu_0}{\pi d}$ d) $\frac{I^2\mu_0}{2\pi d}$

Hints: Resultant force = $F = \sqrt{F_1^2 + F_2^2} = \sqrt{F^2 + F^2} = \sqrt{2F^2} = \sqrt{2}\,F = \sqrt{2}\frac{I^2\mu_0}{2\pi d} = \frac{I^2\mu_0}{\sqrt{2}\pi d}$

9. Which logic operation does the output Q of the above gate combination produce?



a) NOT b) OR **c) AND** d) EXOR

Hints: $\overline{A + B} = \bar{A}.\bar{B} = A.B$

10. In an oscillating LC circuit , the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is:

a) $\frac{Q}{\sqrt{2}}$ b) $\frac{Q}{\sqrt{3}}$ c) $\frac{Q}{2}$ d) Q

Hints: $U_E + U_B = U_{MAX}$; $\frac{q^2}{2C} + \frac{q^2}{2C} = \frac{Q^2}{2C}$; $2q^2 = Q^2$; $q = \frac{Q}{\sqrt{2}}$

11. Which colour of light has the highest speed ?

a) Violet b) red c) green **d) all the above**

12. Duane – Hunt law is :

a) $\lambda = \frac{12400}{V}$ m b) $\lambda = \frac{hc}{V}$ m c) $\lambda = \frac{hc}{\sqrt{E_k}}$ m **d) None of the above**

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13. Two nuclei have mass numbers in the ratio 1:8. Find the ratio of their nuclear radii
a) 1 : 8 **b) 1: 2** c) 2 : 1 d) 8 : 1

Hints: $R = R_0 A^{\frac{1}{3}}$ $R \propto A^{\frac{1}{3}}$ $R_1 : R_2 = \sqrt[3]{1} : \sqrt[3]{8} = 1 : 2$

14. An air core capacitor is charged by a battery. After disconnecting it from the battery dielectric slab is inserted between its plates. Now , which of the following quantities remains constant
a) Energy b) Voltage c) Electric field **d) Charge**

15. The threshold wavelength for a metal surface whose photoelectric work function is 3.313 eV is
a) 2065.5 Å b) 4125 Å c) 6000 Å **d) 3750 Å**

Hints: $W = h\upsilon = \frac{hC}{\lambda_o}$; $\lambda_o = \frac{hC}{W} = \frac{6.626\times10^{-34}\times3\times10^{8}}{3.313\times1.6\times10^{-19}} = \frac{6\times10^{-7}}{1.6} = 3.75\times10^{-7} = 3750Å$

PART – II

Answer any six questions . Question number 24 compulsory. **6 × 2 = 12**

16. What is half life of nucleus? Give its expression.

Half-life($T_{1/2}$) is defined as the time required for the number of atoms initially present to reduce to one half of the initial amount.

$$T_{\frac{1}{2}} = \frac{0.6931}{\lambda}$$

17. What are gravitational waves? How are they produced?

Gravitational waves are the disturbances in the curvature of space – time and it travels with speed of light.

Any accelerated mass emits gravitational waves bit these waves are very weak even for masses like earth.

The strongest source of gravitational waves are black holes.

18. What kind of magnetic properties should (i) permanent magnets and (ii) electromagnets have?

i) Permanent magnets:

The materials with high retentivity, high coercivity and high permeability are suitable for making permanent magnets.

Examples: Steel and Alnico

ii) Electromagnets:

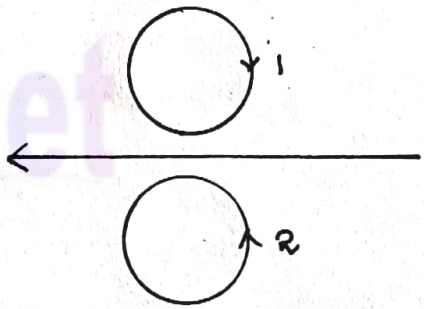
The materials with high initial permeability, low retentivity, low coercivity and thin hysteresis loop with smaller area are preferred to make electromagnets

Example: Soft Iron and Mumetal

19. Using Lenz's law, predict the direction of induced current in conducting rings 1 and 2 when current in the wire is (i)steadily decreasing, and (ii)steadily increasing

(i)steadily decreasing

According to Lenz's law, the induced current opposes the cause of decreasing of magnetic flux. Direction of the current is clockwise in ring 1 and anticlockwise in ring 2.



(ii)steadily increasing

According to Lenz's law, the induced current opposes the cause of increasing of magnetic flux. Direction of the current is clockwise in ring 1 and anticlockwise in ring 2.

20. What are repeaters in an electronic communication system?

Repeaters are used to increase the range or distance through which the signals are sent. It is a combination of transmitter and receiver. The signals are received, amplified, and retransmitted with a carrier signal of different frequency to the destination.

Example: communication satellite in space.

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21. What is the reason of reddish appearance of sun during sunrise and sunset?

During sunrise and sunset, the light from sun travels a greater distance through the atmosphere. Hence, the blue light which has shorter wavelength is scattered away and the less-scattered red light of longer wavelength manages to reach our eye. This is the reason for the reddish appearance of sky during sunrise and sunset

22. The electric field outside a conductor is perpendicular to its surface. Justify

The electric field outside the conductor is perpendicular to the surface of the conductor and has a magnitude of $\frac{\sigma}{\varepsilon_o}$ where σ is the surface charge density at that point.

If the electric field has components parallel to the surface of the conductor, then free electrons on the surface of the conductor would experience acceleration. This means that the conductor is not in equilibrium. Therefore at electrostatic equilibrium, the electric field must be perpendicular to the surface of the conductor.

23. Why electromagnetic waves are called non-mechanical waves?

Electromagnetic waves do not require any medium for propagation. So electromagnetic wave is a non-mechanical wave.

24. A wire of resistance 10Ω is stretched uniformly to thrice its original length. Calculate the resistance of the stretched wire.

$R = \rho\frac{l}{A} = \frac{\rho \times l \times l}{A \times l} = \frac{\rho \times l^2}{V}$ $R \propto l^2$

$R_1 \propto l^2$; $R_2 \propto (3l)^2 \propto 9l^2$ $\frac{R_1}{R_2} = \frac{l^2}{9l^2}$; $R_2 = 9\ R_1 = 9 \times 10 = 90\ \Omega$

PART - III

Answer any six questions . Question number 33 compulsory. **6 × 3 = 18**

25. Find the current through the Zener diode when the load resistance is 1kΩ, Use diode approximation.

Voltage across AB is $V_Z = 9V$

Voltage drop across R = 15 - 9 = 6V

Therefore current through the resistor R, $I = \frac{6}{1\times10^3} = 6$ mA

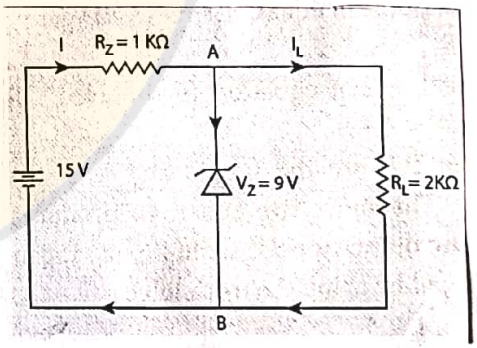
Voltage across the load resistor = $V_{AB} = 9$

Current through load resistor, $I_L = \frac{V_{AB}}{R_L} = \frac{9}{2\times10^3} = 4.5$ mA

The current through the Zener diode, $I_Z = I - I_L$

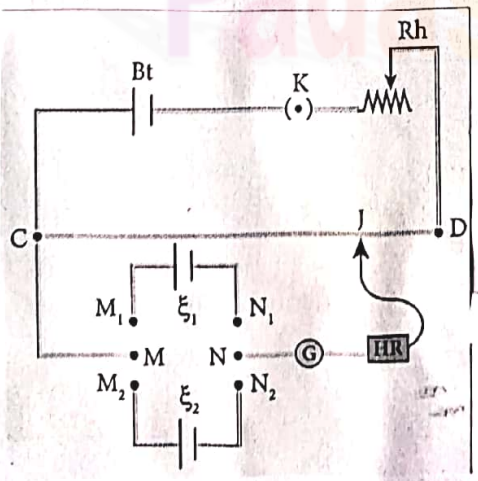
$= 6\text{ mA} - 4.5\text{ mA}$

$= 1.5$ mA



26. How the emfs of two cells are compared using potentiometer?

To compare the emf of two cells, the circuit connections are made as shown in Figure.



Bt - Battery
K - Key
Rh - Rheostat
G - galvanometer
HR - High resistance
J - Jockey.

Potentiometer wire CD is connected to a battery Bt and a key K in series. This is the **primary circuit.**

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The end C of the wire is connected to the terminal M of a DPDT (Double Pole Double Throw) switch and the other terminal N is connected to a jockey through a galvanometer G and a high resistance HR. The cells whose emf ξ_1 and ξ_2 to be compared are connected to the terminals M_1,N_1 and M_2,N_2 of the DPDT switch. The positive terminals of Bt, ξ_1 and ξ_2 should be connected to the same end C. This is the **secondary circuit.**

The DPDT switch is pressed towards M_1, N_1 so that cell ξ_1 is included in the secondary circuit and the balancing length $l1$ is found by adjusting the jockey for zero deflection.

$$\xi_1 = Irl_1 \quad \text{---------- (1)}$$

Then the second cell ξ_2 is included in the circuit and the balancing length l_2 is determined.

$$\xi_2 = Irl_2 \quad \text{---------- (2)}$$

Let r be the resistance per unit length of the potentiometer wire and I be the current flowing through the wire.

By dividing equation (1) by (2) $\frac{\xi_1}{\xi_2} = \frac{l_1}{l_2}$

By including a rheostat (Rh) in the primary circuit, the experiment can be repeated several times by changing the current flowing through it.

27. Derive an expression for de Broglie wavelength of electrons.

. An electron of mass *m* is accelerated through a potential difference of V volt. The kinetic energy acquired by the electron is given by

$$\frac{1}{2}mv^2 = eV$$

Therefore, the speed v of the electron is

$$v = \sqrt{\frac{2eV}{m}}$$

Hence, the de Broglie wavelength of the electron is

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2emV}}$$

Substituting the known values in the above equation, we get

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2V \times 1.6 \times 10^{-11} \times 9 \times 10^{-31}}}$$

$$= \frac{12.27 \times 10^{-10}}{\sqrt{V}} = \frac{12.27}{\sqrt{V}} \text{Å}$$

For example, if an electron is accelerated through a potential difference of 100*V*, then its de Broglie wavelength is 1.227 Å

Since the kinetic energy of the electron,K = eV, then the de Broglie wavelength associated with electron can be also written as $\lambda = \frac{h}{\sqrt{2mK}}$

28. Write notes on moderators.

Moderators: The moderator is a material used **to convert fast neutrons into slow neutrons.** Usually the moderators are chosen in such a way that it must be very light nucleus having mass comparable to that of neutrons. Hence, these light nuclei undergo collision with fast neutrons and the speed of the neutron is reduced (Note that a billiard ball striking a stationary billiard ball of equal mass would itself be stopped but the same billiard ball bounces off almost with same speed when it strikes a heavier mass. This is the reason for using lighter nuclei as moderators).

Most of the reactors use **water, heavy water (D_2O) and graphite** as moderators.

29. A compound microscope has a magnifying power of 100 when the image is formed at infinity. The objective has a focal length of 0.5 cm and the tube length is 6.5 cm. What is the focal length of the eyepiece?

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$m = 100$, $f_o = 0.5$ cm , $L = V_O + f_e = 6.5$ cm , $D = 25$ cm

$$m = \frac{V_0}{u_o} \times \frac{D}{f_e} = -1\left(1 - \frac{V_0}{u_o}\right)\frac{D}{f_e}$$

$$100 = -1\left(1 - \frac{V_0}{0.5}\right)\frac{25}{f_e}$$

$4f_e = -(1 - 2V_o) = -1 + 2V_o$

$2V_o - 4f_e = 1$

$2(6.5 - f_e) - 4f_e = 1$

$13 - 2f_e - 4f_e = 1$

$6f_e = 12$

$f_e = 2$ cm

30. (i) Define current sensitivity and voltage sensitivity of a galvanometer.

Current sensitivity: It is defined as the deflection produced per unit current flowing through it.

Voltage sensitivity: It is defined as the deflection produced per unit voltage applied across it.

(ii) Increasing the current sensitivity does not necessarily increase the voltage sensitivity of a galvanometer. Justify.

Current sensitivity $I_s = \frac{\theta}{I} = \frac{NAB}{K}$ .....(1)

Voltage sensitivity $V_s = \frac{\theta}{IR_g} = \frac{NAB}{KR_g}$ .....(2)

When you increase the number of turns (N) is doubled. The current sensitivity is also doubled from equation (1), but increasing the number of turns correspondingly increases the resistance (K), voltage sensitivity remains unchanged.

Hence increasing the current sensitivity does not necessarily to increase the voltage sensitivity of the galvanometer.

31. Name the power loss that occur during long-range power transmission and give its formula. Explain briefly how this power loss can be overcome.

The electric power generated is transmitted over long distances through transmission lines to reach towns or cities where it is actually consumed. This process is called power transmission. But there is a difficulty during power transmission. A sizable fraction of electric power is lost due to Joule heating $=i^2R$ in the transmission lines which are hundreds of kilometer long. This power loss can be reduced either by reducing current or by reducing resistance of the transmission lines.

The resistance R can be reduced with thick wires of copper or aluminium. But this increases the cost of production of transmission lines and other related expenses. So this way of reducing power loss is not economically viable.

32. Write short notes on (i) Poynting vector, and (ii) intensity of electromagnetic wave.

i) poynting vector

The rate of flow of energy crossing a unit area is known as poynting vector for electromagnetic waves. The poynting vector at any point gives the direction of energy transport from that point.

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) = C^2 \varepsilon_O (\vec{E} \times \vec{B})$$

The unit for poynting vector is **W m^{-2}**

ii) Intensity of the electromagnetic wave

The energy crossing per unit area per unit time and perpendicular to the direction of propagation of electromagnetic wave is called the intensity

$$I = \langle u \rangle C$$

where $\langle u \rangle$ - average energy density, C – velocity of light

(or)

$$I = \frac{\text{total electromagnetic energy(U)}}{\text{surface area (A)} \times \text{time (t)}} = \frac{\text{Power (P)}}{\text{surface area (A)}}$$

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33. Four point charges +q, +q, -q and –q are to be arranged respectively at the four corners of a square PQRS of side r. Find the work needed to assemble this arrangement.

Work done to assemble the charges $U = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r}$

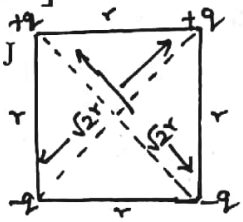
Work done to assemble the charges of the square

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_1q_4}{r_{14}} + \frac{q_2q_3}{r_{23}} + \frac{q_2q_4}{r_{24}} + \frac{q_3q_4}{r_{34}}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{(q)(-q)}{r} + \frac{(q)(-q)}{\sqrt{2}r} + \frac{(q)(-q)}{r} + \frac{(q)(-q)}{r} + \frac{(q)(-q)}{\sqrt{2}r} + \frac{(-q)(-q)}{r}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{-2q^2}{\sqrt{2}r}\right] = 9\times10^9 \times \left(\frac{-\sqrt{2}\,q^2}{r}\right) = -1.273\times10^{10}\ \left(\frac{q^2}{r}\right) \text{J}$$

$$|U| = 1.273 \times 10^{10}\ \left(\frac{q^2}{r}\right) \text{J}$$



PART - IV

Answer all the questions $5 \times 5 = 25$

34. (a) Describe the Fizeau's method to determine speed of light

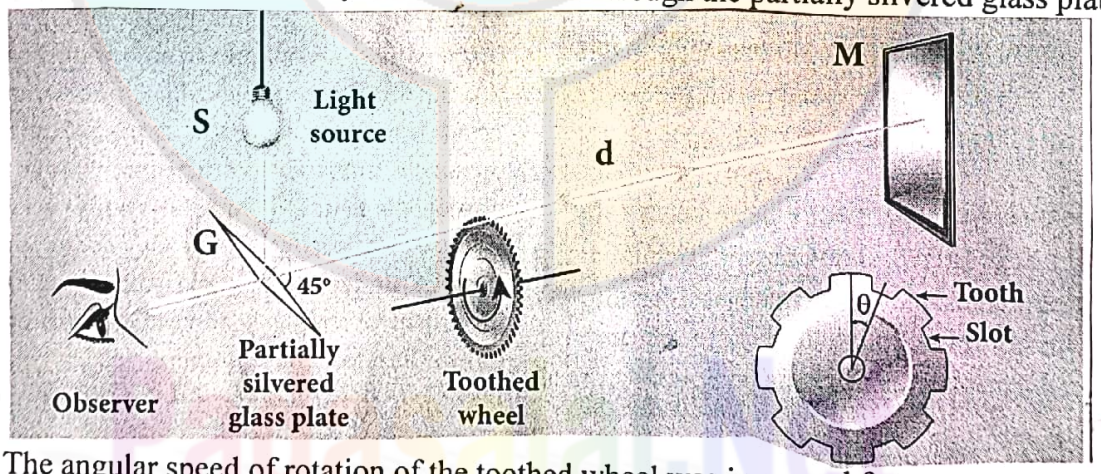
Fizeau's method to determine speed of light:

Apparatus: The apparatus used by Fizeau for determining speed of light in air is shown in Figure. The light from the source S was first allowed to fall on a partially silvered glass plate G kept at an angle of 45° to the incident light from the source.

The light then was allowed to pass through a rotating toothed-wheel with N teeth and N cuts of equal widths whose speed of rotation could be varied through an external mechanism (not shown in the Figure).

The light passing through one cut in the wheel will get reflected by a mirror M kept at a long distance d, about 8 km from the toothed wheel.

If the toothed wheel was not rotating, the reflected light from the mirror would again pass through the same cut and reach the eyes of the observer through the partially silvered glass plate.



Working: The angular speed of rotation of the toothed wheel was increased from zero to value ω until light passing through one cut would completely be blocked by the adjacent tooth. This is ensured by the disappearance of light while looking through the partially silvered glass plate.

Expression for speed of light: The speed of light in air v is equal to the ratio of the distance the light travelled from the toothed wheel to the mirror and back 2d to the time taken t.

$$v = \frac{2d}{t} \quad \text{----------(1)}$$

The distance d is a known value from the arrangement. The time taken t for the light to travel the distance to and fro is calculate from the angular speed ω of the toothed wheel.

$$\omega = \frac{\theta}{t} \quad \text{----------(2)}$$

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The angular speed ω of the toothed wheel when the light disappeared for the first time is, Here, θ is the angle between the tooth and the slot which is rotated by the toothed wheel within that time t.

$$\theta = \frac{\text{total angle of the circle in radian}}{\text{number of teeth+number of cuts}}$$

$$\theta = \frac{2\pi}{2N} = \frac{\pi}{N} \quad \ldots\ldots\ldots\ldots (3)$$

substituting (3) in equation (2), $\omega = \frac{\pi/N}{t} = \frac{\pi}{Nt}$ .....(4)

Rewriting the above equation for t, $t = \frac{\pi}{N\omega}$ .....(5)

Substituting t from equation (5) in equation (1) $v = \frac{2d}{\pi/N\omega}$

After rearranging, $v = \frac{2dN\omega}{\pi}$

The speed of light in air was determined as, $v = 2.99792.10^8$ m s^{-1}.

(OR)

b) Derive the energy expression for hydrogen atom using Bohr atom model.

(i) The energy of an electron in the nth orbit

The electrostatic force is a conservative force, the potential energy for the nth orbit is

$$U_n = \frac{1}{4\pi\varepsilon_o}\frac{(+Ze)(-e)}{r_n} = -\frac{1}{4\pi\varepsilon_o}\frac{Ze^2}{r_n}$$

Substitute $r_n = \frac{\varepsilon_0 h^2}{\pi m e^2}\frac{n^2}{Z}$

$$U_n = \frac{1}{4\varepsilon_0^2}\frac{Z^2 m e^4}{h^2 n^2}\frac{n^2}{Z}$$

The kinetic energy for the nth orbit is

$$KE_n = \frac{1}{2}mv_n^2 = \frac{me^4}{8\varepsilon_0^2 h^2}\frac{Z^2}{n^2}$$

This implies that $U_n = -2\,KE_n$.

Total energy in the nth orbit is

$$E_n = KE_n + U_n = KE_n - 2KE_n = -KE_n$$

$$E_n = = -\frac{me^4}{8\varepsilon_0^2 h^2}\frac{Z^2}{n^2}$$

For hydrogen atom (Z = 1),

$$E_n = -\frac{me^4}{8\varepsilon_0^2 h^2}\frac{1}{n^2} \text{ Joule}$$

where n stands for principal quantum number. The negative sign in the above equation indicates that the electron is bound to the nucleus.

Substituting the values of mass and charge of an electron (m and e), permittivity of free space ε_0 and Planck's constant h and expressing in terms of eV, we get

$$E_n = -13.6\frac{1}{n^2} \text{ eV}$$

(ii) $E_n = \frac{-13.6\ eV}{n^2}$

$$n^2 = \frac{-13.6 \text{ eV}}{-3.4 \text{ e V}} = 4$$

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$n = 2$

$$\text{Angular momentum} = \frac{nh}{2\pi} = \frac{2h}{2\pi} = \frac{h}{\pi} = \frac{6.626 \times 10^{-34}}{3.14} = 2.11 \times 10^{-34} \text{ kg m}^2\text{s}^{-1}$$

35. a) i) State Kirchhoff's current rule and voltage rule

Kirchhoff's first rule (Current rule or Junction rule)

It states that the algebraic sum of the currents at any junction of a circuit is zero.

Kirchhoff's Second rule (Voltage rule or Loop rule)

It states that in a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit

ii) Which material is used for the meter bridge wire and why?

Metre bridge consists of one meter uniform wire of manganin or constantan

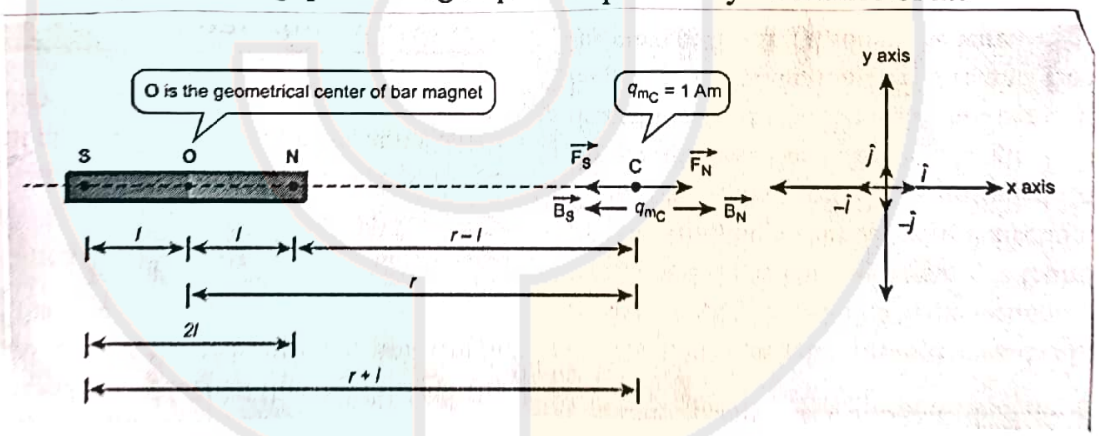
It has high resistivity and low temperature co-efficient of resistance.

The resistance of the wire does not change with increase in temperature of the wire due to flow of current.

(OR)

b) Calculate the magnetic field due to a magnetic dipole (bar magnet) at a point along the axial line.

Consider a bar magnet NS. Let N be the north pole and S be the south pole of the bar magnet. These poles having pole strength q_m and separated by a distance of $2l$.



The magnetic field at a point C (lies along the axis of the magnet) at a distance from the geometrical center O of the bar magnet can be computed by keeping unit north pole (q_{mC} = 1 A m) at C. The force experienced by the unit north pole at C due to pole strength can be computed using Coulomb's law of magnetism as follows:

The force of repulsion between north pole of the bar magnet and unit north pole at point C (in free space) is

$$\vec{F}_N = \frac{\mu_0}{4\pi} \frac{q_m}{(r-l)^2} \hat{i} \quad \text{----------(1)}$$

where $r - l$ is the distance between north pole of the bar magnet and unit north pole at C.

The force of attraction between South Pole of the bar magnet and unit North Pole at point C (in free space) is

$$\vec{F}_S = -\frac{\mu_0}{4\pi} \frac{q_m}{(r+l)^2} \hat{i} \quad \text{----------(2)}$$

where $r + l$ is the distance between south pole of the bar magnet and unit north pole at C.

From equation (1) and (2), the net force at point C is

$$\vec{F} = \vec{F}_N + \vec{F}_S$$

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From definition, this net force is the magnetic field due to magnetic dipole at a point C ($\vec{F} = \vec{B}$)

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q_m}{(r-l)^2}\hat{\imath} + \left(-\frac{\mu_0}{4\pi}\frac{q_m}{(r+l)^2}\hat{\imath}\right)$$

$$\vec{B} = \frac{\mu_0 q_m}{4\pi}\left(\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2}\right)\hat{\imath}$$

$$\vec{B} = \frac{\mu_0\, 2r}{4\pi}\left(\frac{q_m(2l)}{(r^2-l^2)^2}\right)\hat{\imath} \text{ ----------(3)}$$

Since, magnitude of magnetic dipole moment is $|\vec{p}_m| = p_m = q_m.2l$

The magnetic field at a point C equation (3) can be written as $\vec{B} = \frac{\mu_0}{4\pi}\left(\frac{2r\,p_m}{(r^2-l^2)^2}\right)\hat{\imath}$ --------(4)

If the distance between two poles in a bar magnet are small (looks like short magnet) compared to the distance between geometrical centre O of bar magnet and the location of point C i.e., r >>l then, $(r^2 - l^2)^2 \approx r^4$ ----------(5)

Therefore, using equation (5) in equation (4), we get

$$\vec{B} = \frac{\mu_0}{4\pi}\left(\frac{2\,p_m}{r^3}\right)\hat{\imath} = \frac{\mu_0}{4\pi}\frac{2}{r^3}\vec{p}_m$$

where $\vec{p}_m = p_m\hat{\imath}$

36. a) State and prove De Morgan's theorems

The first theorem states that the complement of the sum of two logical inputs is equal to the product of its complements.

Proof

The Boolean equation for NOR gate is $Y = \overline{A+B}$

The Boolean equation for a bubbled AND gate is $Y = \overline{A}.\overline{B}$

A and B are the inputs and Y is the output

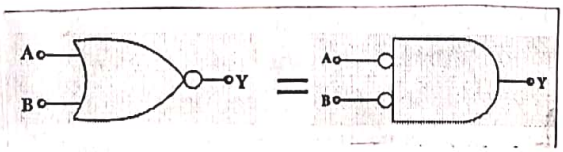
Truth table.

A	B	A + B	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A}.\overline{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

From the above truth table, we can conclude $\overline{A+B} = \overline{A}.\overline{B}$

Thus De Morgan's First Theorem is proved. It also says that a NOR gate is equal to a bubbled AND gate.

The corresponding logic circuit diagram is shown in Figure



De Morgan's Second Theorem

The second theorem states that the complement of the product of two inputs is equal to the sum of its complements.

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Proof

The Boolean equation for NAND gate is $Y = \overline{A.B}$

The Boolean equation for bubbled OR gate is $Y = \overline{A} + \overline{B}$

A and B are the inputs and Y is the output.

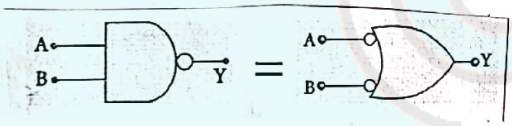
Truth table.

A	B	A . B	$\overline{A.B}$	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

From the above truth table we can conclude $\overline{A.B} = \overline{A} + \overline{B}$

Thus De Morgan's second Theorem is proved. It also says, a NAND gate is equal to a bubbled OR gate.

The corresponding logic circuit diagram is shown in Figure



(OR)

b) i) A transmitter consists of LC circuit with an inductance of 1 μH and a capacitance of 1 μF. What is the wavelength of the electromagnetic waves it emits?

$$f = \frac{1}{2\pi\sqrt{LC}} \quad \ldots\ldots\ldots(1)$$

$$C = f\lambda \; ; \; f = \frac{C}{\lambda} \ldots\ldots\ldots(2)$$

$$\frac{C}{\lambda} = \frac{1}{2\pi\sqrt{LC}} :$$

$$\lambda = C.2\pi\sqrt{LC}$$

$$= 3 \times 10^8 \times 2 \times 3.14 \times \sqrt{10^{-6} \times 10^{-6}} = 18.84 \times 10^2 \text{ m} = 1884 \text{ m}$$

ii) What are Fraunhofer lines ? How are they useful in identifying elements present in the Sun's atmosphere.

When the spectrum obtained from the Sun is examined, it consists of large number of dark lines (line absorption spectrum). These dark lines in the solar spectrum are known as Fraunhofer lines. The absorption spectra for various materials are compared with the Fraunhofer lines in the solar spectrum, which helps in identifying elements present in the Sun's atmosphere.

37 a) Obtain the expression for electric field due to an infinitely long charged wire.

Obtain the expression for electric field due to an infinitely long charged wire.

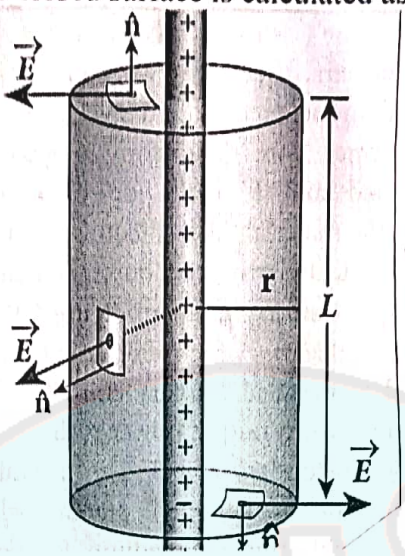
Consider an infinitely long straight wire having uniform linear charge density λ. Let P be a point located at a perpendicular distance r from the wire (Figure1 (a)).

The electric field at the point P can be found using Gauss law. We choose two small charge elements A_1 and A_2 on the wire which are at equal distances from the point P. The resultant electric field due to these two charge elements points radially away from the charged wire and the magnitude of electric field is same at all points on the circle of radius r. This is shown in the Figure 1(b). From this property, we can infer that the charged wire possesses a cylindrical symmetry.

Let us choose a cylindrical Gaussian surface of radius r and length L as shown in the Figure 2.

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The total electric flux in this closed surface is calculated as follows.



$$\Phi_E = \oint \vec{E}.\overrightarrow{dA} = \underset{\text{Curved surface}}{\int} \vec{E}.\overrightarrow{dA} + \underset{\text{top surface}}{\int} \vec{E}.\overrightarrow{dA} + \underset{\text{bottom surface}}{\int} \vec{E}.\overrightarrow{dA} = Q_{encl}/\varepsilon_0 \quad .....(1)$$

It is seen from Figure (2) that for the curved surface,

$$\vec{E}.\overrightarrow{dA} = E.dA.....(2)$$

For the top and bottom surfaces, $\vec{E}$ is perpendicular to $\vec{A}$

$$\vec{E}.\overrightarrow{dA} = 0.....(3)$$

Substituting these values (2) , (3) in the equation (1) and applying Gauss law to the cylindrical surface, we have

$$\Phi_E = \underset{\text{Curved surface}}{\int} EdA = Q_{encl}/\varepsilon_0.....(4)$$

Since the magnitude of the electric field for the entire curved surface is constant, E is taken out of the integration and Q_{encl} is given by

$$Q_{encl} = \lambda L$$

$$E \underset{\text{Curved surface}}{\int} dA = \lambda L/\varepsilon_0.....(5)$$

Here $\int dA$ = total area of the curved surface = $2\pi rL$.

Substituting this in equation (5), we get

$$E.\,2\pi rL = \frac{\lambda L}{\varepsilon_0}$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r} .....(6)$$

In vector form (6) $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r}\hat{r}$

- The electric field due to the infinite charged wire depends on $\frac{1}{r}$
- Equation (6) indicates that the electric field is always along the perpendicular direction ($\hat{r}$) to wire.

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In fact, if $\lambda > 0$ then $\vec{E}$ points perpendicular **outward** ($\hat{r}$) from the wire and if $\lambda < 0$, then $\vec{E}$ points perpendicular **inward .**

- The equation (6) is true only for an **infinitely** long charged wire.

For a charged wire of **finite** length, the electric field need not be radial at all points.

(OR)

b) i) A transmitting antenna has a height of 40 m and the height of the receiving antenna is 30m. What is the maximum distance between them for line-of-sight communication? The radius of the earth is 6.4×10^6 m.

The total distance d between the transmitting and receiving antennas will be the sum of the individual distances of coverage.

$$d = \sqrt{2Rh}$$

$$d = \sqrt{2Rh_1 + 2Rh_2} = \sqrt{2R}\left[\sqrt{h_1} + \sqrt{h_2}\right]$$

$$= \sqrt{2 \times 6.4 \times 10^6}\left[\sqrt{40} + \sqrt{30}\right]$$

$$= \sqrt{2 \times 2 \times 64 \times 10^4 \times 5}\left[\sqrt{40} + \sqrt{30}\right]$$

$$= 2 \times 8 \times 10^2 \times 2.236 \times (6.32 + 5.48)$$

$$= 42217\text{m} = 42.217\text{km}$$

ii) Mention any two advantages of Robotics

Advantages of Robots: (any two)

1) They are much cheaper than humans
2) They never get tired like humans. It can work 24 × 7. Hence absenteeism in work place can be reduced
3) They are more precius and error free in performing the task
4) They are stronger and faster than humans

38. a) Explain the principle , construction and working of transformer.

Principle

The principle of transformer is the mutual induction between two coils. That is, when an electric current passing through a coil changes with time, an emf is induced in the neighbouring coil.

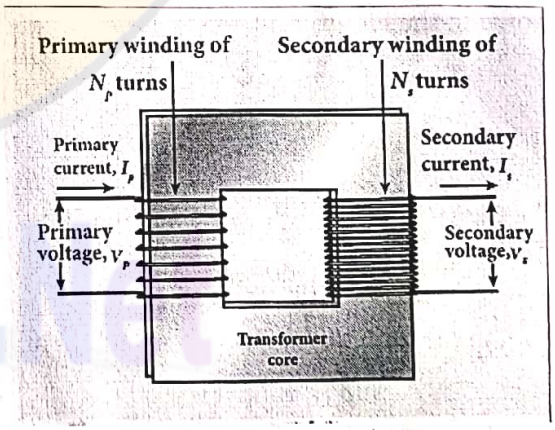
Construction

It consists of two coils of high mutual inductance wound over the same transformer core made up of silicone steel.

To avoid eddy current loss, the core is generally laminated

The alternating voltage is applied across primary coil (P), and the output is taken across secondary coil (S)

The assembled core and coils are kept in a container which is filled with suitable medium for better insulation and cooling purpose.



Working :

The alternating voltage given to the primary coil, set up an alternating magnetic flux in the laminated core.

As the result of flux change, emf is induced in both primary and secondary coils.

The emf induced in the primary coil ε_P is almost equal and opposite to the applied voltage v_p and is given by,

$$v_p = \varepsilon_p = -N_p \frac{d\phi_B}{dt} \text{ ----------(1)}$$

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The frequency of alternating magnetic flux is same as the frequency of applied voltage. Therefore induced in secondary will also have same frequency as that of applied voltage.

The emf induced in the secondary coil ε_s is given by

$$\varepsilon_s = -N_s \frac{d\phi_B}{dt} \text{ ----------(2)}$$

where N_P and N_S ,are the number of turns in the primary and secondary coil.

If the secondary circuit is open, then $\varepsilon_s = v_s$ where v_s is the voltage across secondary coil.

$$v_s = \varepsilon_s = -N_s \frac{d\phi_B}{dt} \text{ ----------(3)}$$

From equations (1) and (3), $\frac{v_s}{v_p} = \frac{N_s}{N_p} = K$ .....(4)

This constant K is known as voltage transformation ratio.

For an ideal transformer, Input power = Output power

$$v_p i_p = v_s i_s$$

where i_p and i_s are the currents in the primary and secondary coil respectively.

Therefore, $\frac{v_s}{v_p} = \frac{N_s}{N_p} = \frac{i_p}{i_s}$ ----------(5)

Equation (5) is written in terms of amplitude of corresponding quantities,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = K$$

i) If $N_s > N_p$ (or $K > 1$), $\therefore V_s > V_p$ and $I_s < I_p$. This is the case of step-up transformer in which voltage is increased and the corresponding current is decreased.

ii) If $N_s < N_p$ (or $K < 1$), $\therefore V_s < V_p$ and $I_s > I_p$ .This is step-down transformer where voltage is decreased and the current is increased.

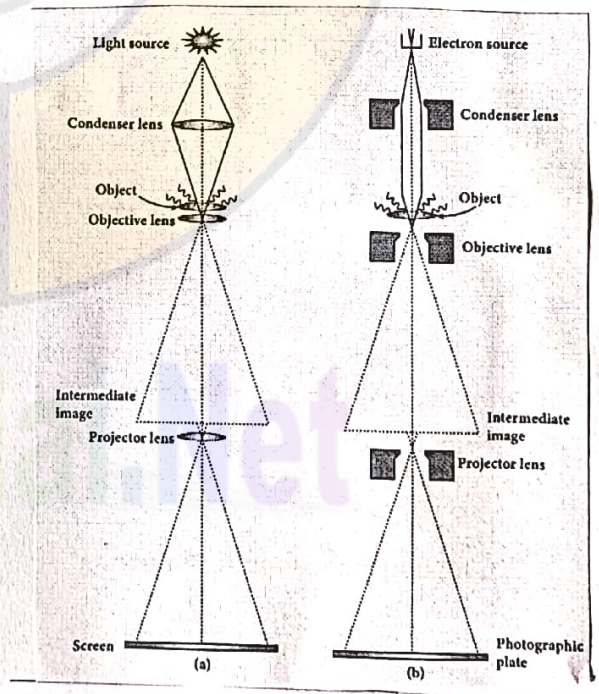
(OR)

b) Explain the principle and working of electron microscope.(Draw the comparison diagrams of optical and electron microscopes

Principle

This is the direct application of wave nature of particles. The wave nature of the electron is used in the construction of microscope called **electron microscope.** The resolving power of a microscope is inversely proportional to the wavelength of the radiation used for illuminating the object under study. Higher magnification as well as higher resolving power can be obtained by employing the waves of shorter wavelengths.

De Broglie wavelength of electron is very much less than (a few thousands less) that of the visible light being used in optical microscopes. As a result, the microscopes employing de Broglie waves of electrons have very much higher resolving power than optical microscope. Electron microscopes giving magnification more than 2,00,000 times are common in research laboratories.



(a) Optical microscope (b) Electron microscope

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Working

The construction and working of an electron microscope is similar to that of an optical microscope except that in electron microscope focussing of electron beam is done by the electrostatic or magnetic lenses. The electron beam passing across a suitably arranged either electric or magnetic fields undergoes divergence or convergence thereby focussing of the beam is done (Figure 7.19).

The electrons emitted from the source are accelerated by high potentials. The beam is made parallel by magnetic condenser lens. When the beam passes through the sample whose magnified image is needed, the beam carries the image of the sample.

With the help of magnetic objective lens and magnetic projector lens system, the magnified image is obtained on the screen.

These electron microscopes are being used in almost all branches of science.

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